

Trialbank
Independent Trial Paper A

2026

Higher School Certificate

Year 12 Trial HSC Examination

Software Engineering

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours 20 minutes
- The HSC exam is completed online; answer on screen in Trialbank, or in the spaces provided on this paper
- Python is used for code questions
- Write your name at the top of every page

Total Marks: 80

Section I
Total marks (20)

- Attempt Questions 1–16
- Allow about 25 minutes for this section

Section II
Total marks (60)

- Attempt ALL questions
- Allow about 1 hour and 55 minutes for this section

STUDENT NAME: _____

Section I

20 marks

Attempt Questions 1–16

Allow about 25 minutes for this section

Questions 1–8: select ONE answer. Questions 9–12: select ALL correct answers (all must be correct for the mark). Questions 13–16: match each item to ONE option; options may be used more than once.

1 **Which line of Python safely queries the database using the variable *name*?**

◯ **A.**
```
cursor.execute("SELECT * FROM users WHERE name = '" + name + "'")
```

◯ **B.**
```
cursor.execute(f"SELECT * FROM users WHERE name = '{name}'")
```

◯ **C.**
```
cursor.execute("SELECT * FROM users WHERE name = ?", (name,))
```

◯ **D.**
```
cursor.execute("SELECT * FROM users WHERE name = %s" % name)
```

2 **A browser requests a page that does not exist on the server. Which HTTP status code is returned?**

◯ **A.** 200

◯ **B.** 301

◯ **C.** 404

◯ **D.** 500

3 **Which statement best describes a progressive web app (PWA)?**

◯ **A.** It can only be installed from an app store

◯ **B.** It runs in a browser but can be installed and work offline using a service worker

◯ **C.** It must be rewritten in native code for each operating system

◯ **D.** It is rendered entirely on the server with no client-side code

4 **A company trains a model on ten years of its past hiring decisions. The model now ranks male applicants higher. What is the most likely cause?**

◯ **A.** Overfitting caused by training for too many epochs

◯ **B.** Bias in the training data

◯ **C.** Underfitting caused by too few features

◯ **D.** A learning rate that is too small

5 **Which OOP concept is best demonstrated by this code?**

```
class Account:
    def __init__(self, balance):
        self.__balance = balance

    def deposit(self, amount):
        if amount > 0:
            self.__balance += amount
```

- ◯ **A.** Inheritance
- ◯ **B.** Polymorphism
- ◯ **C.** Encapsulation
- ◯ **D.** Generalisation

6 **A tester sends thousands of unusual HTTP requests to a running web app to see how it responds. Which method is this?**

- ◯ **A.** Static application security testing (SAST)
- ◯ **B.** Dynamic application security testing (DAST)
- ◯ **C.** Code review
- ◯ **D.** Desk checking

7 **What is the output of this code?**

```
data = [3, 8, 1, 9, 4]
result = [x * 2 for x in data if x > 3]
print(result)
```

- ◯ **A.** [6, 16, 2, 18, 8]
- ◯ **B.** [16, 18, 8]
- ◯ **C.** [8, 9, 4]
- ◯ **D.** [16, 18]

8 **A web page loads slowly because it downloads a 6 MB photo that is displayed 300 pixels wide. What is the best fix?**

- ◯ **A.** Minify the JavaScript files
- ◯ **B.** Resize and compress the image, and serve it in a modern format such as WebP
- ◯ **C.** Add more CSS rules
- ◯ **D.** Serve the page over HTTP instead of HTTPS

9 **Which of the following does HTTPS provide that HTTP does not? Select ALL that apply.**

- ☐ **A.** Data is encrypted in transit
- ☐ **B.** The server's identity is verified using a certificate
- ☐ **C.** SQL injection is prevented
- ☐ **D.** Tampering with data in transit is detected
- ☐ **E.** Pages can load without a server
- ☐ **F.** The domain name is hidden from DNS servers

10 **Select ALL statements that are true for this code.**

```
class Shape:
    def area(self):
        return 0

class Square(Shape):
    def __init__(self, side):
        self.side = side
    def area(self):
        return self.side ** 2

class Circle(Shape):
    def __init__(self, radius):
        self.radius = radius

for s in [Square(3), Circle(2)]:
    print(s.area())
```

- ☐ **A.** Square is a subclass of Shape
- ☐ **B.** Square overrides the area() method
- ☐ **C.** The output is 9 then 12.57
- ☐ **D.** The output is 9 then 0
- ☐ **E.** Circle inherits area() from its superclass
- ☐ **F.** Shape is a subclass of Circle
- ☐ **G.** The loop demonstrates polymorphism

11 **Which of these are supervised learning tasks? Select ALL that apply.**

- ☐ **A.** Predicting a house's price from its size, using past sales
- ☐ **B.** Grouping news articles by topic when none are labelled
- ☐ **C.** Classifying emails as spam using emails already labelled spam or not spam
- ☐ **D.** A robot learning to walk by receiving rewards
- ☐ **E.** Predicting whether a scan is benign using scans labelled by doctors

12 **Which measures reduce the risk or impact of cross-site scripting (XSS)? Select ALL that apply.**

- ☐ **A.** Encoding (escaping) user input before displaying it on a page
- ☐ **B.** Sending a Content Security Policy header
- ☐ **C.** Hashing passwords with bcrypt
- ☐ **D.** Validating and sanitising input
- ☐ **E.** Serving the site over HTTP instead of HTTPS
- ☐ **F.** Setting the HttpOnly flag on session cookies

13 **Match each vulnerability to its description.** **2 marks**

Item	Answer	Options
1. SQL injection		A. An attacker's script runs in another user's browser
2. Cross-site scripting (XSS)		B. A logged-in user's browser is tricked into sending an unwanted request
3. Cross-site request forgery (CSRF)		C. User input changes the meaning of a database query
4. Broken authentication		D. Weak passwords or session handling let an attacker take over an account

14 **Match each machine learning model to the task it suits best.** **2 marks**

Item	Answer	Options
1. Linear regression		A. Grouping customers who have no labels
2. Logistic regression		B. Recognising handwritten digits from pixel values
3. Decision tree		C. Predicting a house price in dollars
4. K-means clustering		D. Predicting the probability that a student passes or fails
5. Neural network		E. Making a loan decision that must be explained as a set of if-then rules

15 **Match each HTTP method to its usual purpose.** **2 marks**

Item	Answer	Options
1. GET		A. Remove a resource
2. POST		B. Retrieve a resource without changing it
3. PUT		C. Submit data to create a new resource
4. DELETE		D. Replace or update an existing resource

16 **Match each activity to the phase of a software engineering project where it mainly belongs.** **2 marks**

Item	Answer	Options
1. Writing the requirements and constraints		A. Identifying and defining
2. Creating a Gantt chart		B. Research and planning
3. Coding modules and reviewing each other's code		C. Producing and implementing
4. Acceptance testing with the client		D. Testing and evaluating
5. Choosing between Agile and Waterfall		
6. Comparing the finished product against the success criteria		

Section II

60 marks

Attempt ALL questions

Allow about 1 hour and 55 minutes for this section

Answer each question in the spaces provided. Write code in Python unless a question says pseudocode is acceptable.

17 Distinguish between symmetric and asymmetric encryption. **2 marks**

18 A login function is shown. **4 marks**

```
def login(username, password):
    query = ("SELECT * FROM users WHERE username = '" + username +
             "' AND password = '" + password + "'")
    return db.execute(query).fetchone()
```

(a) Explain how entering `admin' --` as the username could let an attacker log in without the password. **(2 marks)**

(b) Rewrite the function so it is secure. Assume each user's password is stored as a bcrypt hash in a column called `password_hash`. **(2 marks)**

19 A school is building an app that stores students' names, contact details and timetables. **3 marks**
Describe how the principles of *privacy by design* could be applied to this app.

20 A web app keeps users logged in using a session cookie. **4 marks**

(a) Outline how a session cookie keeps a user logged in between page requests. **(2 marks)**

(b) Describe TWO ways to make the session more secure. **(2 marks)**

21 This Flask route handles a form that adds a score to a leaderboard. **3 marks**

```
@app.route("/scores", methods=["POST"])
def add_score():
    name = request.form["name"]
    score = int(request.form["score"])
    db.execute("INSERT INTO scores (name, score) VALUES (?, ?)",
               (name, score))
    return redirect("/leaderboard")
```

(a) Outline what happens, from browser to database, when the form is submitted. **(2 marks)**

(b) Identify ONE validation check this route is missing. **(1 mark)**

22 A database has these tables: **3 marks**

```
Students(id, name, year)
Results(student_id, subject, mark)
```

Write an SQL query that lists the name and mark of every Year 12 student who scored 80 or more in Software Engineering, highest mark first.

23 Outline TWO ways that following the Web Content Accessibility Guidelines (WCAG) improves a website for its users. **2 marks**

24 Write a Python function `validate_password(pw)` that returns a list of the rules the password fails, in this order: **5 marks**

- `"length"` if it has fewer than 12 characters
- `"digit"` if it has no digit
- `"uppercase"` if it has no uppercase letter
- `"symbol"` if it has none of the characters `!@#$%^&*`

A password that passes every rule returns an empty list. For example, `validate_password("longpassword12")` returns `["uppercase", "symbol"]`.

25 Explain what *overfitting* is, and ONE way to prevent it. **2 marks**

26 A model predicts a student's test score $\hat{y}$ from the hours studied x using the regression line $\hat{y} = 6x + 40$. **4 marks**

(a) Predict the score of a student who studies for 7 hours. **(1 mark)**

(b) Explain why the model is unreliable for a student who studies for 20 hours. **(2 marks)**

(c) The model is tested on three students: (2 hours, 50), (5 hours, 72) and (8 hours, 86). Calculate the mean squared error. **(1 mark)**

27 Evaluate the impact of automation through machine learning on employment and on society. **6 marks**

28 A single neuron has inputs $x_1 = 2$ and $x_2 = 3$, weights $w_1 = 0.5$ and $w_2 = -1$, a bias of 1 and a ReLU activation function, where ReLU(z) = max(0, z). **4 marks**

(a) Calculate the neuron's output. Show your working. **(2 marks)**

(b) Outline how training would change this neuron if its output was too low. **(2 marks)**

29 A library system stores items. Every **LibraryItem** has a `title` and an `item_id`, and a method `describe()` that returns a string. A **Book** is a LibraryItem that also has an `author`. A **DVD** is a LibraryItem that also has a `runtime` in minutes. Book and DVD each override `describe()` to include their extra attribute. **5 marks**

Write Python code for all three classes.

30 With reference to your code in the previous question, explain how it shows *generalisation*, *inheritance* and *polymorphism*. Use the terms superclass and subclass. **3 marks**

31 A team is building a canteen ordering app using Scrum. **4 marks**

(a) Write a user story for ordering lunch, with TWO acceptance criteria. **(2 marks)**

(b) Explain how working in sprints helps the team manage changing requirements. **(2 marks)**

32 A school canteen app lets students log in, order food and pay online. Justify FOUR security measures the developers should build into the app, covering different parts of the system. **6 marks**

End of paper